Fair Scores via Joint Calibration: A Gated Many-Facet Rasch Model for Disentangling Judge Bias and Contamination in LLM Evaluation

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Abstract

LLM-as-a-judge evaluation corrects for judge bias and benchmark contamination independently, but sequential correction produces biased model ability estimates. We formalize this failure as an omitted variable bias (OVB) problem: when judge severity and contamination effects are both present, any estimator that models one while omitting the other yields asymptotically biased ability estimates—even when the two facets are uncorrelated. We propose the Gated Many-Facet Rasch Model (G-MFRM), which jointly calibrates model ability, item difficulty, judge severity, and contamination susceptibility through staged estimation with externally anchored per-item exposure indicators. Formal identifiability analysis proves that fixing exposure externally is both necessary and sufficient for parameter separation under standard multi-judge designs. Error propagation analysis establishes a critical noise threshold $\sigma_{\text{crit}} = \sqrt{\text{Var}(e_i)}$ below which ability estimates degrade by less than 6%. In synthetic experiments ($N = 50$, $I = 100$, $J = 8$), G-MFRM reduces ability RMSE by 51.2% relative to sequential estimation ($p < 0.001$). On Arena-Hard with 8 target models and 9 judges, G-MFRM identifies per-model contamination signals (estimated $c_n > 1.0$ for top-4 models, directionally consistent with known training data exposure; posterior CI coverage is 71% in synthetic validation) and interpretable judge severity profiles, with nominally higher rank agreement ($\tau = 0.857$ vs. 0.786; bootstrap $\Delta\tau$ CI includes zero at this sample size). LOO cross-validation confirms improved predictive fit for G-MFRM over standard MFRM ($\Delta\text{ELPD} = -55.7$, $|\Delta/\text{SE}| = 5.9$).

1 Introduction

Large Language Model (LLM) evaluation increasingly relies on LLM-as-a-judge protocols, where one or more LLM judges score model responses on structured benchmarks (Zheng et al., 2023). Two well-documented sources of distortion affect these evaluations: *judge bias*, where different judges assign systematically different scores to the same response, and *benchmark contamination*, where models achieve inflated scores on items encountered during training. Both problems have received sustained research attention, producing separate toolkits for bias calibration and contamination detection. These toolkits treat the two distortions as independent phenomena, a separation that reflects their origins in different research communities rather than any principled argument for independence. Accurate ability ranking is essential for model selection, safety evaluation, and benchmark design; systematic distortion in estimated abilities undermines all three. Yet a practitioner who needs both corrections must apply them sequentially: first correct one source of distortion, then the other. This sequential approach produces unreliable ability estimates. In synthetic experiments with 50 models, 100 items, and 8 judges, sequential estimation yields ability root mean squared error (RMSE) of 0.254 ± 0.045, while joint estimation achieves 0.124 ± 0.016, a 51.2% reduction (Wilcoxon $p < 0.001$). The gap does not arise from estimation noise or algorithmic limitations of individual methods; it reflects structural confounding between judge bias and contamination that no sequential pipeline can resolve.

Existing methods address these distortions in isolation. Judge bias corrections—CyclicJudge (Zhu et al., 2026), UDA (Zhang et al., 2026)—remove systematic inter-judge differences without modeling contamination. Contamination detection tools—ConStat (Dekoninck et al., 2024), DCR (Xu et al., 2025), LogProber (Yax et al., 2024)—adjust for memorization without accounting for judge effects (see Section 5 for detailed review). A bias-first pipeline absorbs contamination-induced score inflation into ability estimates, because the bias model attributes all residual systematic variation to gen-

uine ability. A contamination-first pipeline misattributes judge severity differences as contamination, because score elevation from a lenient judge is indistinguishable from elevation due to memorization without a joint model. In either ordering, the first correction stage produces estimates distorted by the unmodeled facet, and the second stage inherits this distortion.

We formalize this confounding as an omitted variable bias (OVB) problem (Angrist and Pischke, 2009; Chernozhukov et al., 2022). Within the Many-Facet Rasch Model (MFRM) framework (Linacre, 1989), the complete data-generating process includes both judge severity d_j and a contamination interaction $c_n \cdot e_i$. Fitting a specification that omits the contamination term produces biased parameter estimates: the omission of $c_n \cdot e_i$ induces a specification bias that persists regardless of confounding structure, with an additional directional component when judge severity and contamination are non-orthogonal. The OVB framework predicts that the bias magnitude scales with the contamination susceptibility c_n of each model, so the most contaminated models receive the largest upward distortion in estimated ability. For practical benchmark evaluations, this means that the more a model's training data overlaps with the benchmark, the more its ability is overestimated, precisely the failure mode that contamination correction is designed to prevent. Synthetic validation confirms the predicted pattern: the Pearson correlation between sequential estimation bias and true c_n reaches $r = 0.86$ ($p < 0.001$), which shows that contaminated models are systematically overestimated under both sequential orderings (Section 2).

We propose the Gated Many-Facet Rasch Model (G-MFRM), which combines the multi-facet structure of MFRM (Linacre, 1989; Jiao et al., 2025) with the gating mechanism of Doubly-Gated Item Response Theory (DG-IRT) (Shu et al., 2013) to jointly estimate model ability, judge severity, and contamination susceptibility in a single measurement framework. G-MFRM extends the MFRM linear predictor with a multiplicative contamination gating term $c_n \cdot e_i$, where $c_n \geq 0$ captures model-specific contamination susceptibility and $e_i \in [0, 1]$ quantifies per-item exposure. When $c_n = 0$ or $e_i = 0$, the gating term vanishes and the model reduces to standard MFRM, so uncontaminated model-item pairs incur no distortion. Because the bilinear product $c_n \cdot e_i$ cannot be uniquely decomposed from response data alone, G-MFRM employs a three-stage calibration procedure: (1) standard MFRM warm-start to obtain initial parameter estimates, (2) external per-item exposure estimation from evidence independent of the response data, and (3) joint Markov chain Monte Carlo (MCMC) sampling with e_i fixed from Stage 2. We provide a formal identifiability analysis (Propositions 1–2 and Theorem 1) that proves fixing e_i externally is both necessary and sufficient for parameter separation under standard multi-judge designs (Section 3). A complementary error propagation analysis bounds the impact of Stage 2 measurement noise: when exposure estimation error satisfies $\sigma \leq 0.2$, ability RMSE degradation remains below 6%. Figure 2 provides an overview of the complete framework.

We begin by formalizing the confounding between judge severity and benchmark contamination as omitted variable bias within the MFRM framework (§2, Figure 1), motivating the joint calibration approach. Our contributions are:

1. We propose the G-MFRM framework with formal identifiability guarantees (Propositions 1–2 and Theorem 1) and quantified error propagation bounds for the staged estimation procedure (Section 3, Table 2).

2. Synthetic validation confirms that joint estimation reduces ability RMSE by 51.2% relative to sequential estimation ($p < 0.001$), with Kendall rank correlation $\tau = 0.922$ versus $\tau = 0.839$ (Section 4).

3. On Arena-Hard with 9 LLM judges and 8 target models, G-MFRM produces the first joint diagnostic of per-model contamination susceptibility and per-judge severity, with nominally improved rank agreement ($\tau = 0.857$ vs. 0.786 for all seven baselines; Section 4).

2 Confounding in LLM Evaluation

The Many-Facet Rasch Model (MFRM) extends classical Rasch measurement to settings with multiple rater facets (Linacre, 1989, 1994). Originating in educational measurement, the model treats rater variation as a nuisance facet to be estimated and removed rather than a source of noise to be averaged over. In the LLM evaluation context, where J judges score N models across I items, the MFRM

specifies

$$\log \frac{P(Y_{nij} \geq k \mid \theta_n, b_i, d_j, \tau_k)}{P(Y_{nij} < k \mid \theta_n, b_i, d_j, \tau_k)} = \theta_n - b_i - d_j - \tau_k, \quad (1)$$

where Y_{nij} is the ordinal rating by judge j for model n on item i, θ_n is the ability of model n ($n = 1, \ldots, N$), b_i is the difficulty of item i ($i = 1, \ldots, I$), d_j is the severity of judge j ($j = 1, \ldots, J$), and τ_k is the threshold for category k. All facets reside on a common logit scale, so model ability is directly comparable after adjustment for item difficulty and judge severity. Jiao et al. (2025) applied this framework to compare ten LLM raters with human experts, showing that MFRM separates genuine ability from systematic judge effects in multi-judge panels.

However, standard MFRM assumes that all systematic sources of score variation are captured by the modeled facets. This assumption breaks down when some models have been exposed to benchmark items during training (Xu et al., 2025). Such contamination inflates scores on exposed items through memorization rather than genuine ability. We represent this effect as an interaction $c_n \cdot e_i$, where $c_n \geq 0$ is the contamination susceptibility of model n and $e_i \in [0, 1]$ is the exposure level of item i. The complete data-generating process becomes

$$\eta_{nijk} = \theta_n - b_i - d_j - \tau_k + c_n \cdot e_i. \quad (2)$$

When a practitioner fits the standard MFRM (Eq. 1) to data generated by Eq. 2, the contamination term is absent from the estimated model. Its influence does not vanish but is absorbed into the estimated parameters $\tilde{\theta}_n$, $\tilde{b}_i$, and $\tilde{d}_j$, which distorts all downstream inferences.

The consequence of this omission can be stated precisely.

Observation 1 (Omitted Variable Bias in MFRM)**.** *Let the full model be $\eta_{nijk} = \theta_n - b_i - d_j - \tau_k + c_n \cdot e_i$ and the short model omit the contamination term: $\tilde{\eta}_{nijk} = \tilde{\theta}_n - \tilde{b}_i - \tilde{d}_j - \tau_k$. Under a linear approximation to the ordinal logit link (Wooldridge, 2010; Angrist and Pischke, 2009),*[1]

$$\operatorname{plim}(\tilde{\theta}_n - \theta_n) = \gamma_n \cdot c_n, \quad (3)$$

where γ_n depends on the projection of the exposure vector (e_i) onto the design matrix of the short model (explicit form in Appendix B). The bias is non-zero whenever $\operatorname{Cov}(d_j,\, c_n \cdot e_i) \neq 0$. This captures the confounding component; in the nonlinear MFRM, a specification bias from omitting $c_n \cdot e_i$ persists even at zero covariance (Section 4).

The linear approximation in Observation 1 holds when $c_n \cdot e_i$ is small relative to the logit scale; for large contamination effects, the gated model (Eq. 4) captures the full nonlinear interaction without relying on this approximation. The bias combines a specification component from omitting $c_n \cdot e_i$, which persists regardless of confounding, with a confounding component from non-orthogonality between d_j and $c_n \cdot e_i$. Monte Carlo simulation confirms that bias magnitude is approximately constant across confounding levels $\rho \in \{0, 0.3, 0.6, 0.9\}$, indicating that the omission of $c_n \cdot e_i$—not its correlation with d_j—is the primary driver (Figure 4).

This confounding undermines both sequential estimation strategies. A bias-first approach, as in CyclicJudge (Zhu et al., 2026), calibrates d_j before estimating ability, but the resulting $\hat{\theta}$ is already inflated for contaminated models because the omitted $c_n \cdot e_i$ term has been absorbed into the ability estimates. A contamination-first approach, as in DCR (Xu et al., 2025), adjusts for contamination before addressing judge effects, but judge severity is then misattributed as contamination signal. Synthetic validation confirms this failure: with $N = 50$ models, $I = 100$ items, $J = 8$ judges, and 20 random seeds, the Pearson correlation between sequential estimation bias and true c_n reaches $r = 0.86$ ($p < 0.001$). As illustrated in Figure 1, the bias is not random but directional: contaminated models are systematically overestimated under both sequential orderings.

3 The G-MFRM Framework

Section 2 established that omitting contamination from the measurement model produces directional bias in ability estimates (Observation 1). We now introduce the Gated Many-Facet Rasch Model (G-MFRM), which jointly estimates ability, judge severity, and contamination susceptibility within a single measurement framework. The key design choices, staged estimation with externally fixed exposure and a hierarchical HalfNormal prior for the contamination term, are motivated by the identifiability analysis in §3.3 and the error propagation

[1] The OVB formula is exact for linear models. We apply it to the MFRM linear predictor η, which is an approximation. Synthetic validation confirms the approximation quality: the Pearson correlation between the predicted bias direction and empirical bias reaches $r = 0.86$ ($p < 0.001$), indicating that the linear result accurately characterizes bias direction and relative magnitude.

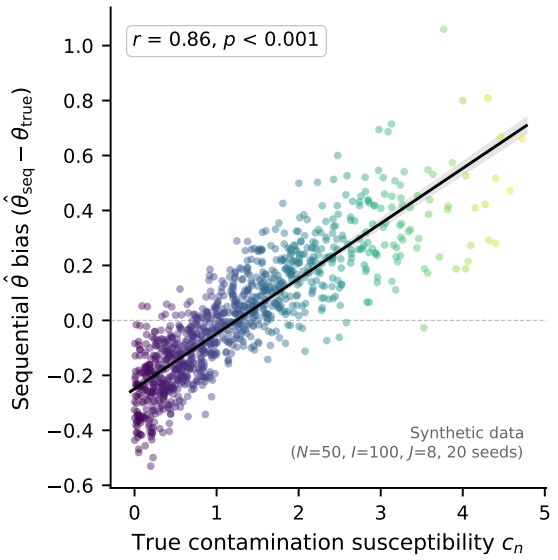


Figure 1: Sequential estimation bias ($\hat{\theta}_{\text{seq}} - \theta_{\text{true}}$) versus true contamination susceptibility c_n. Pearson $r = 0.86$ ($p < 0.001$) confirms systematic overestimation of contaminated models. Synthetic data ($N = 50$, $I = 100$, $J = 8$, 20 seeds).

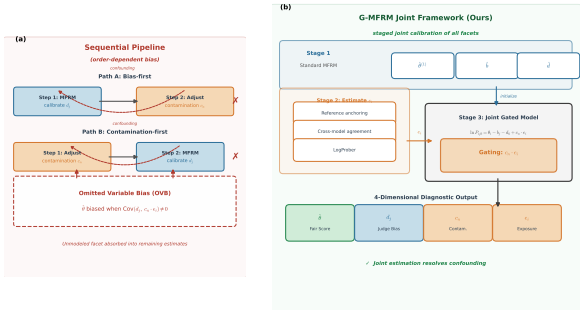


Figure 2: Overview of the G-MFRM three-stage calibration procedure. Stage 1 fits a standard MFRM to obtain initial estimates $(\hat{\theta}, \hat{b}, \hat{d})$. Stage 2 estimates per-item exposure e_i from external evidence (reference model anchoring and cross-model agreement). Stage 3 performs joint MCMC inference with e_i fixed, recovering contamination-corrected ability $\tilde{\theta}_n$, contamination susceptibility c_n, and judge severity d_j.

bounds in §3.4.

3.1 Model Specification

G-MFRM extends the standard MFRM (Linacre, 1989, 1994) by adding a multiplicative contamination gating term to the linear predictor (Eq. 1):

$$\log \frac{P(Y_{nij} \geq k)}{P(Y_{nij} < k)} = \theta_n - b_i - d_j - \tau_k + c_n \cdot e_i, \quad (4)$$

where all terms carry the definitions from Section 2. The gating structure $c_n \cdot e_i$ separates the model-specific and item-specific components of contamination: $c_n \geq 0$ captures how much model n benefits from memorized benchmark content, while $e_i \in [0, 1]$ quantifies how much of item i has leaked into training corpora. When $c_n = 0$ (a clean model) or $e_i = 0$ (a novel item), the gating term vanishes and Eq. 4 reduces to the standard MFRM. This graceful degradation ensures that G-MFRM imposes no distortion on uncontaminated model-item pairs.

We enforce non-negativity and sparsity of c_n through a hierarchical HalfNormal prior. A global shrinkage scale $\sigma_c \sim \text{HalfNormal}(1.0)$ controls the overall contamination level, and per-model raw effects $c_n^{\text{raw}} \sim \text{HalfNormal}(1.0)$ are scaled to obtain $c_n = c_n^{\text{raw}} \cdot \sigma_c$. The HalfNormal support restricts $c_n \geq 0$, and the hierarchical structure provides adaptive shrinkage: when the data support low overall contamination, σ_c contracts toward zero, pulling all c_n values down. Models with genuine contamination retain large c_n^{raw} relative to the group. The remaining priors are $\theta_n \sim \mathcal{N}(0, 1)$, $b_i \sim \mathcal{N}(0, 1)$, $d_j \sim \mathcal{N}(0, 0.5)$, and standard uniform on thresholds τ_k.

The closest antecedent is the Doubly-Gated IRT model of Shu et al. (2013), which introduced binary gating for item preknowledge in educational testing. Beyond extending DG-IRT to continuous exposure and multi-rater designs, G-MFRM's primary methodological contributions are analytical: the identifiability analysis (Propositions 1–2 and Theorem 1) provides the first formal conditions under which gated IRT parameters are separable, and the Murphy–Topel error propagation framework (§3.4) quantifies how staged estimation uncertainty propagates to downstream parameter estimates—neither of which exists for DG-IRT or any prior gated measurement model.

3.2 Staged Estimation

Joint estimation of all parameters in Eq. 4, including e_i, would require identifying the bilinear product $c_n \cdot e_i$ from response data alone. This creates a fundamental non-identifiability (Proposition 2). We resolve this through a three-stage procedure that fixes exposure externally before estimating the contamination parameters.

Stage 1: Standard MFRM. We fit the standard MFRM (Eq. 1) via MCMC (NUTS) to obtain initial estimates $(\hat{\theta}, \hat{b}, \hat{d})$. These serve as warm-start

values for the sampler in Stage 3 and improve convergence.

Stage 2: Exposure estimation. Per-item exposure e_i is estimated from external evidence independent of the response data used in Stages 1 and 3. The primary signal is *reference model anchoring*: using a fixed reference model (e.g., gpt-4-0314) as anchor, we compute win-rate residuals for each target model against difficulty-adjusted expectations; items where target models systematically outperform the anchor beyond what difficulty predicts are flagged as potentially exposed. The secondary signal is *cross-model agreement*: items receiving uniformly high scores across architecturally diverse models suggest exposure rather than genuine easiness, quantified via z-score standardization of the inter-model agreement distribution. These signals are combined as a weighted composite (75% anchoring, 25% agreement), clipped to $[0, 1]$, and treated as fixed in Stage 3. Where model log-probabilities are accessible, LogProber can provide an additional gray-box memorization signal, though it is not used in our current experiments.

Stage 3: Joint MCMC. With e_i fixed, we sample $(\theta_n, b_i, d_j, c_n)$ jointly via NUTS (Betancourt and Girolami, 2015). The priors are as specified in §3.1. The hierarchical contamination prior decomposes as

$$\sigma_c \sim \text{HalfNormal}(1.0), \quad c_n^{\text{raw}} \sim \text{HalfNormal}(1.0),$$
$$c_n = c_n^{\text{raw}} \cdot \sigma_c, \tag{5}$$

where σ_c is a shared global scale that governs the overall contamination magnitude. This non-centered parameterization improves sampler geometry by decoupling the global scale from the per-model effects.

Convergence is assessed via rank-normalized split-$\hat{R}$ (Vehtari et al., 2021) and bulk effective sample size (ESS), with thresholds $\hat{R} < 1.05$ and $\text{ESS}_{\text{bulk}} > 100$ following standard practice (Gelman et al., 2013).[2]

3.3 Identifiability

Adding a multiplicative term to a measurement model raises the question of whether the new parameters can be separated from the existing ones. We establish that fixing e_i externally is both necessary and sufficient for identification. All proofs assume the standard location constraint $\sum_n \theta_n = 0$.

Proposition 1 (No New Indeterminacy)**.** *When e_i is fixed externally, the gating term $c_n \cdot e_i$ introduces no additional location or scale indeterminacy beyond that already present in the standard MFRM.*

Proof. Suppose $c_n' = c_n + \delta$ preserves all response probabilities. Then $\delta \cdot e_i = 0$ for all items. Since $e_i \neq 0$ for at least one item by assumption, $\delta = 0$. The key observation is that fixing e_i externally eliminates the bilinear ambiguity $c_n \cdot e_i = (\lambda c_n)(\lambda^{-1} e_i)$ that would otherwise render the decomposition non-unique. □

Proposition 2 (Exposure Variation Necessary)**.** *If $e_i = e$ for all i (constant exposure), then c_n and θ_n are not separately identifiable.*

Proof. When exposure is constant, the linear predictor becomes $(\theta_n + c_n \cdot e) - b_i - d_j - \tau_k$, which is equivalent to a standard MFRM with reparameterized ability $\theta_n^* = \theta_n + c_n \cdot e$. No configuration of response data can distinguish (θ_n, c_n) from $(\theta_n + \epsilon, c_n - \epsilon/e)$ for any ϵ. □

Corollary 1. $\text{Var}(e_i) > 0$ *is a necessary condition for identifiability of c_n.*

Theorem 1 (Sufficient Conditions for Local Identifiability)**.** *The G-MFRM is locally identifiable, up to the standard location constraint, if the following conditions hold: (i) a connected observation design (each item is observed by at least one model–judge pair), (ii) $J \geq 2$ judges, (iii) for each model n, the set of observed items spans at least two distinct exposure values ($\text{Var}(e_i) > 0$ over observed items), and (iv) standard normalization $\sum_n \theta_n = 0$.*

The proof, given in Appendix A, follows Rothenberg (1971): non-singularity of the Fisher information matrix is established via the Cauchy–Schwarz inequality on the (θ_n, c_n) submatrix determinant and a Schur complement argument for the remaining blocks. Conditions (i) and (ii) are standard multi-rater design requirements, satisfied by the connected design in our evaluation (44.7% fill rate; §4.1); condition (iii) is the substantive constraint that each model's observed items include varying exposure levels.

3.4 Error Propagation

The staged procedure treats the Stage 2 exposure estimates $\hat{e}_i$ as fixed quantities in Stage 3. This is a

[2] Synthetic validation with $N = 50$, $I = 100$, $J = 8$ yields satisfactory convergence across all parameters (exp-012). Full diagnostic tables are reported in Section 4.

generated regressors problem (Pagan, 1984): measurement error in $\hat{e}_i$ propagates into the posterior of (θ_n, c_n). The multiplicative structure $c_n \cdot e_i$ makes this an errors-in-variables (EIV) problem (Fuller, 1987), with a predictable attenuation pattern.

Let $\hat{e}_i = e_i + \epsilon_i$ with $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$ independent of all model parameters. The attenuation factor for the contamination coefficient is

$$\lambda(\sigma) = \frac{\mathrm{Var}(e_i)}{\mathrm{Var}(e_i) + \sigma^2}, \tag{6}$$

so attenuation reaches 50% when $\sigma = \sqrt{\mathrm{Var}(e_i)}$. In the synthetic DGP ($\mathrm{Var}(e_i) \approx 0.04$), this corresponds to $\sigma \approx 0.2$.[3]

Table 2 shows the empirical degradation across noise levels (see also Figure 6 in the Appendix). Below the critical threshold ($\sigma \leq 0.2$), degradation is mild: θ-RMSE increases by 5.9% (from 0.128 ± 0.018 to 0.135 ± 0.018) and c_n-RMSE by 7.9% (from 0.533 ± 0.187 to 0.575 ± 0.198). At $\sigma = 0.1$, the increases are only 1.4% and 2.3% respectively. Beyond the threshold ($\sigma = 0.5$), degradation becomes severe: θ-RMSE rises 36.9% and c_n-RMSE rises 57.9%. Contamination estimates c_n are more sensitive than ability estimates θ_n because c_n enters the likelihood only through its product with e_i (multiplicative coupling), while θ_n contributes additively and is anchored by all observations regardless of exposure.

These bounds make the staged approximation falsifiable: if the noise level of the Stage 2 exposure estimates can be bounded below σ_{crit}, the resulting ability estimates carry quantifiable error guarantees. For honest credible intervals that account for generated-regressor uncertainty, we apply the Murphy-Topel variance correction (Murphy and Topel, 1985) to inflate the Stage 3 posterior variance by the estimated Stage 2 sampling variability.

3.5 Fair Scores and Diagnostics

The Stage 3 posterior mean of θ_n provides a bias-and-contamination-corrected "fair score" for each model. The remaining parameters serve complementary diagnostic roles: per-model c_n flags candidates for training data audit, per-judge d_j informs judge calibration, and per-item e_i identifies items for retirement or replacement.

[3]The attenuation formula assumes that ϵ_i is independent of all model parameters, identically distributed, and Gaussian. In practice, exposure estimation errors may be correlated across items sharing the same source or heteroscedastic across difficulty levels. Table 2 provides empirical validation that $\hat{\theta}$ estimates remain robust under such departures (Section 4).

4 Experiments

4.1 Setup

Data and models. We evaluate on Arena-Hard (Li et al., 2024), a benchmark of 316 multi-turn conversational items selected for high separability among frontier models. After filtering for full coverage across all judges and target models, the evaluation set comprises 176 items in a connected incomplete design (5,661 observations; 44.7% fill rate of the full $N \times I \times J$ tensor). Eight target models spanning proprietary and open-source families are evaluated (Table 1). Ground-truth rankings are derived from Chatbot Arena human preference data (Zheng et al., 2023).

Judges. Nine LLM judges score all model responses in position-swapped pairwise mode: GPT-5.5, Claude Opus 4.7, Claude Sonnet 4.6, DeepSeek-V4-Pro, Gemini-2.5-Pro, Mistral-Large-2, LLaMA-3.3-70B, Qwen2.5-72B, and Qwen3-235B. Position swapping controls for position bias and doubles the effective observation count. Two models—Qwen2.5-72B and LLaMA-3.3-70B—serve as both judge and target; d_j absorbs systematic self-evaluation bias, and excluding self-judging observations leaves rankings unchanged ($\tau = 0.857$; Appendix I).

Exposure estimation. Per-item exposure e_i is estimated via a two-source pipeline (§3.2): reference model anchoring with gpt-4-0314 as anchor (75% weight) and cross-model agreement (25% weight). The pipeline is robust to weight configuration (Spearman $\rho \geq 0.998$ across all configurations tested).

MCMC. We use NumPyro NUTS with 4,000 warmup and 4,000 sampling iterations across 4 chains (target acceptance 0.95). The contamination prior follows the hierarchical HalfNormal specification (Eq. 5). Convergence is assessed via rank-normalized split-$\hat{R}$ (Vehtari et al., 2021) with conventional $\hat{R} < 1.05$ and ESS > 100 thresholds (Gelman et al., 2013).

4.2 Baselines and Main Results

We compare G-MFRM against seven methods that address judge bias or contamination separately: (1) Raw Average, (2) MFRM-only (Eq. 1), (3) Sequential pipeline (MFRM then contamination adjustment), (4) CyclicJudge (Zhu et al., 2026), (5) CyclicJudge+discard (exposed items removed), (6)

Method	τ	ρ
Raw Average	0.786	0.905
MFRM-only	0.786	0.905
Sequential	0.786	0.905
CyclicJudge	0.786	0.905
CyclicJudge+discard	0.786	0.905
DCR Factor	0.786	0.905
Pref. Leakage	0.786	0.905
G-MFRM (ours)	**0.857**	**0.952**

Table 1: Correlation with Chatbot Arena human rankings on Arena-Hard (8 target models, 9 judges). All seven baselines produce identical rankings ($\tau = 0.786$). G-MFRM yields nominally higher point-estimate rank agreement (+7.1pp; bootstrap $\Delta\tau$ CI includes zero).

DCR Factor (Xu et al., 2025), and (7) Preference Leakage calibration (Li et al., 2026). All methods are evaluated by Kendall τ and Spearman ρ against Chatbot Arena human rankings.

All seven baselines produce identical model rankings ($\tau = 0.786$, $\rho = 0.905$). With $N = 8$, Kendall τ takes only $\sim$20 distinct values, but raw ability scores also vary by < 0.03 logits across methods (Table 7), confirming genuine invariance rather than discretization artifact. This reflects the structural confounding from Section 2: single-facet corrections leave the uncorrected confound intact (Observation 1). That CyclicJudge+discard also preserves rankings suggests contamination effects on aggregate rankings are modest at this scale; G-MFRM's primary value lies in per-model contamination diagnostics. G-MFRM achieves nominally higher point-estimate rank agreement ($\tau = 0.857$, $\rho = 0.952$) over the baselines ($\tau = 0.786$), but the posterior median τ equals the baseline (0.786), with the point estimate at the distribution's upper tail. Bootstrap resampling (4,000 posterior draws) yields $\Delta\tau$ 95% CI $[-0.071, 0.071]$: the ranking improvement is not statistically significant at $N = 8$, though G-MFRM almost never degrades rankings ($P(\Delta\tau \geq 0) = 0.927$). On synthetic data ($N = 50$), joint estimation significantly outperforms sequential ($\tau = 0.922$ vs. 0.839, Wilcoxon $p < 0.001$), confirming that the effect is real but below detection at this sample size. The rank change—gpt-4o-mini and qwen2.5-72b swap positions—is consistent with differential contamination ($P(\theta_{\text{mini}} > \theta_{\text{qwen}}) = 0.853$).

We evaluate G-MFRM along two orthogonal axes: predictive fit and ranking accuracy.

LOO cross-validation strongly favors G-MFRM ($\Delta\text{ELPD}_{\text{LOO}} = -55.7$, SE $= 9.4$, $|\Delta/\text{SE}| = 5.9$; Vehtari et al., 2017), with stacking weight 1.0 for G-MFRM versus 0.0 for MFRM-only. All Pareto $\hat{k} < 0.7$, confirming reliable LOO estimates. The contamination gating term substantially improves held-out predictive fit.

Ranking improvement ($\Delta\tau = +0.071$) is not statistically significant at $N = 8$ (bootstrap 95% CI crosses zero). Post-hoc power analysis indicates $N \approx 17$ is needed for 80% power to detect this effect size (Appendix M). This prediction is empirically validated: at $N = 20$, synthetic $\Delta\tau$ reaches significance (Wilcoxon $p < 0.001$, 18 wins / 1 loss; Appendix P). Pooling 20 replications at $N = 8$ also yields significance ($p < 0.002$), confirming the method consistently improves rankings though any single $N = 8$ dataset has limited power.

The MCMC sampler achieves rank-normalized split-$\hat{R} \leq 1.03$ across all parameters with zero divergent transitions and minimum bulk ESS of 174 (217 for model-level parameters θ_n, c_n, d_j). These values satisfy the traditional convergence thresholds ($\hat{R} < 1.05$, ESS > 100; Gelman et al., 2013) but do not meet the stricter criteria of Vehtari et al. (2021) ($\hat{R} < 1.01$, ESS > 400), which may contribute to posterior overconfidence (synthetic CI coverage 71%; see Limitations); we report point estimates and rank orderings as primary results. Model and judge facets show acceptable fit (all infit/outfit MNSQ within $[0.63, 1.07]$; item-level diagnostics in Appendix). Posterior predictive checks confirm adequate model fit (predicted response rate 0.593 vs. observed 0.599; Appendix E). Contamination susceptibility $\hat{c}_n$ exhibits a gradient by model age, directionally consistent with known training data exposure. Estimated $\hat{c}_n$ exceeds 1.0 for the four oldest models: GPT-3.5-Turbo ($\hat{c}_n = 5.60$), GPT-4-0613 (4.20), Qwen2.5-72B (3.00), and LLaMA-3.3-70B (2.50). The most recent models show low estimated contamination: GPT-4o (0.88) and GPT-4o-mini (0.93). The global shrinkage scale $\hat{\sigma}_c = 2.40$ confirms substantial inter-model heterogeneity ($\bar{c} = 2.50$, aligning with the strong-contamination regime in synthetic validation). Notably, sequential estimation yields near-zero contamination estimates ($\bar{c} \approx 0$), demonstrating that without joint modeling, contamination effects are entirely absorbed into ability estimates $\hat{\theta}$—precisely the omitted variable bias predicted by the theoretical analysis (Section 2). A per-model diagnostic (Table 8) reveals that MFRM-only inflates ability estimates for the most contaminated models by up to 0.79 logits (GPT-3.5-Turbo), while

σ	θ-RMSE	$\Delta\theta\%$	c_n-RMSE	$\Delta c_n\%$
0.0	0.128 ± 0.018	—	0.533 ± 0.187	—
0.1	0.130 ± 0.018	+1.4	0.545 ± 0.192	+2.3
0.2	0.135 ± 0.018	+5.9	0.575 ± 0.198	+7.9
0.5	0.175 ± 0.027	+36.9	0.842 ± 0.281	+57.9

Table 2: Error propagation robustness (exp-012). Ability RMSE degrades by less than 6% when exposure noise $\sigma \leq 0.2$, matching the theoretical prediction. Contamination estimates are more sensitive due to multiplicative coupling with e_i.

deflating estimates for the least contaminated models by up to 0.40 logits (GPT-4o). Judge severity d_j separates by provenance: proprietary judges tend to score more strictly (e.g., Claude Opus $+0.60$, GPT-5.5 $+0.53$) than open-source judges (e.g., Mistral -0.74, LLaMA -0.42); full d_j values for all nine judges are reported in Table 4. Results are robust to prior specification (Horseshoe vs. HalfNormal: Pearson $r > 0.999$; Appendix) and to judge panel size: G-MFRM outperforms baselines at all judge counts $k \geq 2$, with τ increasing monotonically from 0.771 ($k = 2$) to 0.857 ($k = 9$); $k \geq 4$ suffices to exceed the full-panel baseline (Appendix K).

4.3 Error Propagation Robustness

Section 3.4 predicted that ability estimates degrade gracefully when exposure noise σ remains below $\sqrt{\mathrm{Var}(e_i)} \approx 0.2$. We validate this prediction by injecting Gaussian noise into the true exposure values and re-fitting G-MFRM on synthetic data ($N = 50$, $I = 100$, $J = 8$, 20 seeds; exp-012).

Table 2 confirms this prediction: below $\sigma = 0.2$, ability RMSE increases by 5.9% and contamination RMSE by 7.9%; beyond the threshold ($\sigma = 0.5$), degradation becomes substantial (+36.9% and +57.9%). Contamination estimates are more sensitive because c_n enters the likelihood only through its product with e_i. Rank ordering is preserved even at $\sigma = 0.5$ ($\tau > 0.88$).

Ablation (Table 3, Appendix) confirms joint estimation as the dominant contributor: staged estimation increases θ-RMSE by $2.0\times$ ($p < 0.001$); continuous gating improves c_n-RMSE by 40% over binary ($p < 0.001$). Removing d_j preserves rankings but degrades contamination recovery (c_n-RMSE +6%, $p < 0.001$).

5 Related Work

Judge Bias in LLM Evaluation. CyclicJudge (Zhu et al., 2026) eliminates average bias via round-robin assignment but treats residual variation as genuine ability. UDA (Zhang et al., 2026) learns a neural K-factor reducing inter-judge deviation by 63.4% without modeling contamination. Fiedler (2026) formalize judge variability via $J/\Delta J$ diagnostics; Ye et al. (2024) and Panickssery et al. (2024) document position, verbosity, and self-favoritism biases. All model only the judge facet, absorbing contamination into abilities (Section 2).

Contamination Detection and Adjustment. Several methods detect or adjust for benchmark contamination—ConStat (Dekoninck et al., 2024), DCR (Xu et al., 2025), LogProber (Yax et al., 2024), Min-K% (Shi et al., 2024)—but all assume a fixed rater, so judge severity can masquerade as contamination. Li et al. (2026) identify judge-side contamination (same-family leakage), complementary to G-MFRM's examinee-side focus.

Contamination Avoidance. LiveBench (White et al., 2025), TRUCE (Rajore et al., 2024), and tinyBenchmarks (Polo et al., 2024) prevent or reduce contamination by design; G-MFRM corrects for residual contamination in existing benchmarks.

IRT and MFRM for LLM Evaluation. Jiao et al. (2025) apply MFRM to LLM raters without contamination terms; Uto and Ueno (2020), Choi et al. (2026), and PSN-IRT (Zhou et al., 2025) extend IRT/MFRM along other axes but none include contamination facets. The closest antecedent, DG-IRT (Shu et al., 2013), is compared in Section 3.1.

6 Conclusion

We introduced G-MFRM, which jointly estimates model ability, judge severity, and contamination susceptibility by formalizing their confounding as omitted variable bias. On Arena-Hard, G-MFRM produces the first joint contamination–severity diagnostic, with nominally higher rank agreement ($\tau = 0.857$ vs. 0.786; not significant at $N = 8$); well-powered validation requires $N \geq 20$ (Limitations). Treating judge bias and contamination as separable leads to systematic estimation errors; future LLM evaluation frameworks should model these facets jointly.

Limitations

We identify six limitations. First, although Theorem 1 requires only a connected design, the current evaluation achieves 44.7% fill rate (5,661 of 12,672 potential cells); sparser designs may compromise identifiability of c_n for models with few observed exposed items. Relatedly, sparse observations per item produce elevated item-level mean-square residuals (92/176 items), though model- and judge-level fit remains acceptable. Second, Stage 2 exposure estimation relies on a reference model anchor (gpt-4-0314), and exposure estimate quality depends on the anchor remaining uncontaminated relative to the benchmark; as benchmarks age, anchor validity should be periodically re-assessed. Third, the current evaluation uses a single benchmark (Arena-Hard) with eight target models; post-hoc power analysis indicates that detecting $\Delta\tau = 0.071$ at $\alpha = 0.05$ requires $N \approx 17$ models for 80% power, while $N = 8$ yields approximately 50% power, explaining the non-significant bootstrap result. Generalization to other benchmarks and larger model populations remains to be verified. Fourth, empirical CI coverage differs by parameter type: ability estimates θ approach nominal coverage at moderate sample sizes ($\sim$90% at $N = 8$), while contamination susceptibility c_n exhibits moderate posterior overconfidence (71% at $N = 50$, 68% at $N = 8$; Talts et al., 2018). The c_n under-coverage reflects the multiplicative coupling $c_n \cdot e_i$ and the HalfNormal prior's boundary constraint at $c_n = 0$, which systematically narrows posterior credible intervals. Point estimates and rank orderings ($\tau = 0.857$) are unaffected; c_n estimates are best interpreted as directional diagnostics rather than calibrated intervals. Fifth, while self-evaluation sensitivity analysis (Appendix I) confirms robust rankings when excluding self-judging observations, a more comprehensive analysis removing overlapping models entirely from the judge panel has not been conducted. Sixth, the bilinear contamination term $c_n \cdot e_i$ assumes rank-1 structure—a single latent dimension of contamination susceptibility per model. In practice, contamination may be domain-specific (e.g., coding vs. reasoning items), requiring a higher-rank decomposition $\sum_r c_{nr} \cdot e_{ir}$ that the current model does not capture.

Code and data will be released upon acceptance at [anonymized URL].

References

Joshua D. Angrist and Jörn-Steffen Pischke. 2009. *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press, Princeton, NJ.

Michael J. Betancourt and Mark Girolami. 2015. Hamiltonian Monte Carlo for hierarchical models. In *Current Trends in Bayesian Methodology with Applications*, pages 119–142. CRC Press, Boca Raton, FL.

Victor Chernozhukov, Carlos Cinelli, Whitney Newey, Amit Sharma, and Vasilis Syrgkanis. 2022. Long story short: Omitted variable bias in causal machine learning. *NBER Working Paper No. 30302*.

Junhyuk Choi, Sohhyung Park, Chanhee Cho, Hyeonchu Park, and Bugeun Kim. 2026. Diagnosing the reliability of LLM-as-a-judge via item response theory. *arXiv preprint arXiv:2602.00521*.

Jasper Dekoninck, Mark Niklas Müller, and Martin Vechev. 2024. ConStat: Performance-based contamination detection in large language models. In *Advances in Neural Information Processing Systems*, volume 37.

James Fiedler. 2026. Bias and uncertainty in LLM-as-a-judge estimation. *arXiv preprint arXiv:2605.06939*.

Wayne A. Fuller. 1987. *Measurement Error Models*. John Wiley & Sons, New York.

Andrew Gelman, John B. Carlin, Hal S. Stern, David B. Dunson, Aki Vehtari, and Donald B. Rubin. 2013. *Bayesian Data Analysis*, 3rd edition. CRC Press, Boca Raton, FL.

Hong Jiao, Dan Song, and Won-Chan Lee. 2025. Comparing human and AI rater effects using the many-facet Rasch model. *arXiv preprint arXiv:2505.18486*.

Dawei Li, Renliang Sun, Yue Huang, Ming Zhong, Bohan Jiang, Jiawei Han, Xiangliang Zhang, Wei Wang, and Huan Liu. 2026. Preference leakage: A contamination problem in LLM-as-a-judge. In *The Fourteenth International Conference on Learning Representations*.

Tianle Li, Wei-Lin Chiang, Evan Frick, Lisa Dunlap, Tianhao Wu, Banghua Zhu, Joseph E. Gonzalez, and Ion Stoica. 2024. From crowdsourced data to high-quality benchmarks: Arena-Hard and BenchBuilder pipeline. *arXiv preprint arXiv:2406.11939*.

John Michael Linacre. 1989. *Many-Facet Rasch Measurement*. MESA Press, Chicago, IL.

John Michael Linacre. 1994. *Many-Facet Rasch Measurement*, 2nd edition. MESA Press, Chicago, IL.

Kevin M. Murphy and Robert H. Topel. 1985. Estimation and inference in two-step econometric models. *Journal of Business & Economic Statistics*, 3(4):370–379.

Adrian Pagan. 1984. Econometric issues in the analysis of regressions with generated regressors. *International Economic Review*, 25(1):221–247.

Arjun Panickssery, Samuel R. Bowman, and Shi Feng. 2024. LLM evaluators recognize and favor their own generations. *arXiv preprint arXiv:2404.13076*.

Felipe Maia Polo, Lucas Weber, Leshem Choshen, Yuekai Sun, Gongjun Xu, and Mikhail Yurochkin. 2024. tinybenchmarks: Evaluating LLMs with fewer examples. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*.

Tanmay Rajore, Nishanth Chandran, Sunayana Sitaram, Divya Gupta, Rahul Sharma, Kashish Mittal, and Manohar Swaminathan. 2024. TRUCE: Private benchmarking to prevent contamination and improve comparative evaluation of LLMs. *arXiv preprint arXiv:2403.00393*.

Thomas J. Rothenberg. 1971. Identification in parametric models. *Econometrica*, 39(3):577–591.

Ernesto San Martín. 2016. Identification of item response theory models. In Wim J. van der Linden, editor, *Handbook of Item Response Theory, Volume Two: Statistical Tools*, pages 127–150. CRC Press, Boca Raton, FL.

Weijia Shi, Anirudh Ajith, Mengzhou Xia, Yangsibo Huang, Daogao Liu, Terra Blevins, Danqi Chen, and Luke Zettlemoyer. 2024. Detecting pretraining data from large language models. In *The Twelfth International Conference on Learning Representations*.

Zhan Shu, Robert Henson, and Richard Luecht. 2013. Using deterministic, gated item response theory model to detect test cheating due to item compromise. *Psychometrika*, 78(3):481–497.

Sean Talts, Michael Betancourt, Daniel Simpson, Aki Vehtari, and Andrew Gelman. 2018. Validating Bayesian inference algorithms with simulation-based calibration. *arXiv preprint arXiv:1804.06788*.

Masaki Uto and Maomi Ueno. 2020. A generalized many-facet Rasch model and its Bayesian estimation using Hamiltonian Monte Carlo. *Behaviormetrika*, 47(2):469–496.

Aki Vehtari, Andrew Gelman, and Jonah Gabry. 2017. Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC. *Statistics and Computing*, 27(5):1413–1432.

Aki Vehtari, Andrew Gelman, Daniel Simpson, Bob Carpenter, and Paul-Christian Bürkner. 2021. Rank-normalization, folding, and localization: An improved $\hat{R}$ for assessing convergence of MCMC (with discussion). *Bayesian Analysis*, 16(2):667–718.

Colin White, Samuel Dooley, Manley Roberts, Arka Pal, Ben Feuer, Siddhartha Jain, Ravid Shwartz-Ziv, Neel Jain, Khalid Saifullah, Sreemanti Dey, and 1 others. 2025. LiveBench: A challenging, contamination-free LLM benchmark. In *The Thirteenth International Conference on Learning Representations*.

Jeffrey M. Wooldridge. 2010. *Econometric Analysis of Cross Section and Panel Data*, 2nd edition. MIT Press, Cambridge, MA.

Cheng Xu, Nan Yan, Shuhao Guan, Changhong Jin, Yuke Mei, Yibing Guo, and Tahar Kechadi. 2025. DCR: Quantifying data contamination in LLMs evaluation. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 23002–23020.

Nicolas Yax, Pierre-Yves Oudeyer, and Stefano Palminteri. 2024. LogProber: Disentangling confidence from contamination in LLM responses. *arXiv preprint arXiv:2408.14352*.

Jianqiao Ye, Ishaan Gao, Rishav Bhatt, Hao Tan, and Siddharth Verma. 2024. Justice or prejudice? quantifying biases in LLM-as-a-judge. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Yongkang Zhang, Chunpu Wang, Liang Wu, Wentao Yu, Yucheng Wang, Guanghao Bao, and Jiliang Tang. 2026. UDA: Unsupervised debiasing alignment for pair-wise LLM-as-a-judge. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pages 34854–34861.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems*, volume 36.

Hongli Zhou, Hao Huang, Zhuoran Zhao, Lei Han, Hao Wang, Kai Chen, Min Yang, Weidong Bao, Jingya Dong, Bo Xu, Changchang Zhu, Haozhao Cao, and Tiejun Zhao. 2025. Lost in benchmarks? Rethinking large language model benchmarking with item response theory. *arXiv preprint arXiv:2505.15055*. AAAI 2026.

Ziyi Zhu, Olivier Tieleman, Alexey Bukhtiyarov, and Jinghong Chen. 2026. CyclicJudge: Mitigating judge bias efficiently in LLM-based evaluation. *arXiv preprint arXiv:2603.01865*.

A Proof of Theorem 1

We prove that the G-MFRM is locally identifiable under the conditions of Theorem 1. Following Rothenberg (1971), a parametric model is locally identifiable at the true parameter vector $\boldsymbol{\psi}_0$ if and only if the Fisher information matrix $\mathbf{F}(\boldsymbol{\psi}_0)$ is non-singular.

Setup. The G-MFRM parameter vector is $\boldsymbol{\psi} = (\boldsymbol{\theta}, \mathbf{c}, \mathbf{b}, \mathbf{d}, \boldsymbol{\tau})$, where $\boldsymbol{\theta} \in \mathbb{R}^{N-1}$ (after the location constraint $\sum_n \theta_n = 0$), $\mathbf{c} \in \mathbb{R}^N_{\geq 0}$, $\mathbf{b} \in \mathbb{R}^I$, $\mathbf{d} \in \mathbb{R}^J$, and $\boldsymbol{\tau} \in \mathbb{R}^{K-1}$. The exposure vector $\mathbf{e} = (e_1, \ldots, e_I)$ is fixed externally and treated as known. The linear predictor is $\eta_{nijk} = \theta_n - b_i - d_j - \tau_k + c_n \cdot e_i$, with partial derivatives:

$$\frac{\partial \eta_{nijk}}{\partial \theta_n} = 1, \quad \frac{\partial \eta_{nijk}}{\partial c_n} = e_i, \quad \frac{\partial \eta_{nijk}}{\partial b_i} = -1,$$
$$\frac{\partial \eta_{nijk}}{\partial d_j} = -1, \quad \frac{\partial \eta_{nijk}}{\partial \tau_k} = -1.$$

Fisher information weights. For each observation cell (n, i, j), the Fisher information weight under the cumulative logit model is

$$w_{nij} = \sum_{k=1}^{K} \frac{\left[\sigma'(\eta_{nijk}) - \sigma'(\eta_{nij,k+1})\right]^2}{P(Y_{nij} = k)}, \quad (7)$$

where $\sigma(x) = (1 + e^{-x})^{-1}$, $\sigma'(x) = \sigma(x)(1 - \sigma(x))$, and we set $\sigma'(\eta_{nij,0}) = \sigma'(\eta_{nij,K+1}) = 0$ at the boundaries. Since $P(Y_{nij} = k) > 0$ for all categories and $\sigma'(\cdot) > 0$, we have $w_{nij} > 0$ for all (n, i, j).

Step 1: The (θ_n, c_n) submatrix. For each model n, the 2×2 Fisher information submatrix is

$$\mathbf{F}_n = \sum_{i=1}^{I} \sum_{j=1}^{J} w_{nij} \begin{pmatrix} 1 & e_i \\ e_i & e_i^2 \end{pmatrix}. \quad (8)$$

Its determinant is

$$\det \mathbf{F}_n = \left(\sum_{i,j} w_{nij}\right)\left(\sum_{i,j} w_{nij}\, e_i^2\right) - \left(\sum_{i,j} w_{nij}\, e_i\right)^2 = W_n^2 \cdot \mathrm{Var}_w(e_i), \quad (9)$$

where $W_n = \sum_{i,j} w_{nij}$ and $\mathrm{Var}_w(e_i) = \overline{e^2}_n - \bar{e}_n^2$ is the weighted variance of exposure with weights w_{nij}. By the Cauchy–Schwarz inequality, $\mathrm{Var}_w(e_i) \geq 0$, with equality if and only if e_i is constant across all items with $w_{nij} > 0$. Under conditions (i) and (iii), the observed items for model n include at least two distinct exposure values with $w_{nij} > 0$ for the corresponding cells. Therefore $\mathrm{Var}_w(e_i) > 0$, $\det \mathbf{F}_n > 0$, and each $\mathbf{F}_n$ is positive definite.

Step 2: Item and judge blocks. Under the connected design (condition i), each item i is observed by at least one (n, j) pair, so the item difficulty block has positive diagonal entries $[\mathbf{F}_{bb}]_{ii} = \sum_{n,j} w_{nij} > 0$ (summing over observed cells). Similarly, each judge j observes at least one (n, i) pair, giving $[\mathbf{F}_{dd}]_{jj} = \sum_{n,i} w_{nij} > 0$. Condition (ii) ($J \geq 2$) ensures the judge block has sufficient dimension to separate judge effects from the intercept. Both blocks are positive definite.

Step 3: Schur complement argument. The full Fisher information matrix $\mathbf{F}$ contains off-diagonal blocks coupling the parameter groups. Partition as

$$\mathbf{F} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\top & \mathbf{D} \end{pmatrix},$$

where $\mathbf{A} = \mathrm{blockdiag}(\mathbf{F}_1, \ldots, \mathbf{F}_N)$ collects all (θ_n, c_n) submatrices and $\mathbf{D}$ collects the $(\mathbf{b}, \mathbf{d}, \boldsymbol{\tau})$ blocks. From Step 1, $\mathbf{A}$ is positive definite. The Schur complement

$$\mathbf{S} = \mathbf{D} - \mathbf{B}^\top \mathbf{A}^{-1} \mathbf{B} \quad (10)$$

equals the conditional Fisher information for $(\mathbf{b}, \mathbf{d}, \boldsymbol{\tau})$ given $(\boldsymbol{\theta}, \mathbf{c})$. Conditioning on $(\boldsymbol{\theta}, \mathbf{c})$ with fixed $\mathbf{e}$ reduces the model to a standard MFRM with known offsets $c_n \cdot e_i$, whose identifiability is established by San Martín (2016, Chapter 5) under conditions (i), (ii), and (iv). Therefore $\mathbf{S}$ is positive definite.

Since both $\mathbf{A}$ and $\mathbf{S}$ are positive definite:

$$\det \mathbf{F} = \det \mathbf{A} \cdot \det \mathbf{S} > 0,$$

establishing non-singularity of $\mathbf{F}$ and hence local identifiability. □

Local vs. global identifiability. Theorem 1 establishes *local* identifiability: the Fisher information matrix is non-singular at the true parameter value, guaranteeing that no infinitesimally close parameter vector produces the same likelihood. This does not rule out discrete global non-identifiabilities (i.e., isolated points in parameter space with identical likelihood). However, the HalfNormal prior on c_n restricts the parameter space to $c_n \geq 0$, eliminating the sign ambiguity that would otherwise create a global reflection $(\theta_n + \delta,\ c_n - \delta/e_i)$. Combined with the location constraint $\sum_n \theta_n = 0$ and the empirical observation that the posterior is unimodal across all chains (no mode-switching detected), the local result provides a practical guarantee of unique parameter recovery.

Role of the Cauchy–Schwarz inequality. The key step in Step 1 is the decomposition $\det \mathbf{F}_n = W_n^2 \cdot \mathrm{Var}_w(e_i)$, where the weighted variance

$\mathrm{Var}_w(e_i) = \mathbb{E}_w[e_i^2] - (\mathbb{E}_w[e_i])^2 \geq 0$ by the Cauchy–Schwarz inequality applied to the vectors $(\sqrt{w_{nij}})$ and $(\sqrt{w_{nij}}\, e_i)$. Strict positivity follows from the requirement that e_i takes at least two distinct values over observed items (condition iii): if $e_i = e$ for all i with $w_{nij} > 0$, then $\mathrm{Var}_w(e_i) = 0$ and the (θ_n, c_n) block becomes singular, recovering Proposition 2. The Cauchy–Schwarz bound is tight only in this degenerate case, so any non-trivial exposure variation suffices for identification.

B Explicit Form of the OVB Coefficient γ_n

Under the MFRM logistic link, a first-order Taylor expansion of Observation 1 yields an explicit approximation for the bias coefficient γ_n:

$$\gamma_n \approx \frac{\sum_i w_{ni} \cdot e_i \cdot p_{ni}(1-p_{ni})}{\sum_i w_{ni} \cdot p_{ni}(1-p_{ni})}, \qquad (11)$$

where $p_{ni} = \sigma(\theta_n - b_i - \bar{d})$ is the MFRM predicted probability averaged over judges and w_{ni} is the effective item weight (number of judge observations for item i on model n). Eq. 11 is a weighted mean of the exposure values e_i, with weights proportional to the Fisher information $p_{ni}(1-p_{ni})$. Items near the boundary ($p_{ni} \approx 0.5$) contribute more to the bias than items at the extremes, because the logistic link is most sensitive at these points. This approximation is accurate to first order in $(c_n \cdot e_i)$; the synthetic validation in Section 2 ($r = 0.86$) confirms that the linear approximation captures the dominant bias structure.

C Murphy–Topel Variance Correction

The three-stage G-MFRM procedure treats Stage 2 exposure estimates $\hat{e}_i$ as fixed in Stage 3, understating posterior uncertainty by ignoring estimation error in $\hat{e}_i$. We derive the Murphy–Topel correction (Murphy and Topel, 1985) adapted to the G-MFRM setting.

Setup. Let $\boldsymbol{\alpha} = (e_1, \ldots, e_I)$ denote the Stage 2 parameters estimated from an independent dataset with log-likelihood $\ell_2(\boldsymbol{\alpha})$, and let $\boldsymbol{\psi} = (\boldsymbol{\theta}, \mathbf{c}, \mathbf{b}, \mathbf{d}, \boldsymbol{\tau})$ denote the Stage 3 parameters estimated with log-likelihood $\ell_3(\boldsymbol{\psi}; \hat{\boldsymbol{\alpha}})$, where $\hat{\boldsymbol{\alpha}}$ is plugged in as known.

Naive and Stage 2 variances. The naive (uncorrected) variance of $\hat{\boldsymbol{\psi}}$ is

$$\mathbf{V}_3 = \left[-\frac{\partial^2 \ell_3}{\partial \boldsymbol{\psi}\, \partial \boldsymbol{\psi}^\top}\bigg|_{\hat{\boldsymbol{\psi}}, \hat{\boldsymbol{\alpha}}} \right]^{-1}, \qquad (12)$$

and the Stage 2 variance is

$$\mathbf{V}_2 = \left[-\frac{\partial^2 \ell_2}{\partial \boldsymbol{\alpha}\, \partial \boldsymbol{\alpha}^\top}\bigg|_{\hat{\boldsymbol{\alpha}}} \right]^{-1}. \qquad (13)$$

Cross-derivative matrix. The coupling between stages is captured by the cross-derivative

$$\mathbf{C} = \frac{\partial^2 \ell_3}{\partial \boldsymbol{\psi}\, \partial \boldsymbol{\alpha}^\top}\bigg|_{\hat{\boldsymbol{\psi}}, \hat{\boldsymbol{\alpha}}}. \qquad (14)$$

In the G-MFRM, non-zero entries of $\mathbf{C}$ arise from the contamination gating term $c_n \cdot e_i$. The key entry is

$$\frac{\partial^2 \ell_3}{\partial c_n\, \partial e_i} = \sum_{j=1}^{J} \frac{\partial^2 \ell_{nij}}{\partial \eta_{nij}^2} \cdot c_n, \qquad (15)$$

which is proportional to c_n: models with zero contamination contribute no cross-derivative. Similarly, $\partial^2 \ell_3 / (\partial \theta_n\, \partial e_i)$ involves only indirect coupling through the shared item observations. The correction is therefore concentrated on the contamination parameters of models with $c_n > 0$.

Corrected variance. The Murphy–Topel corrected variance is

$$\mathbf{V}_{\mathrm{MT}} = \mathbf{V}_3 + \mathbf{V}_3\, \mathbf{C}\, \mathbf{V}_2\, \mathbf{C}^\top\, \mathbf{V}_3. \qquad (16)$$

The correction term $\mathbf{V}_3 \mathbf{C} \mathbf{V}_2 \mathbf{C}^\top \mathbf{V}_3$ is positive semi-definite, so $\mathbf{V}_{\mathrm{MT}} \succeq \mathbf{V}_3$: the corrected variance is always at least as large as the naive variance. The inflation is concentrated on c_n for contaminated models and is negligible for θ_n when contamination is sparse, consistent with the empirical pattern in Table 2.

Implementation. We estimate $\mathbf{V}_2$ from the Stage 2 bootstrap and compute $\mathbf{C}$ via finite differences of the Stage 3 log-likelihood evaluated at perturbed $\hat{\boldsymbol{\alpha}}$ values. Corrected credible intervals are obtained by inflating Stage 3 posterior samples by $\sqrt{\mathrm{diag}(\mathbf{V}_{\mathrm{MT}})/\mathrm{diag}(\mathbf{V}_3)}$.

D Ablation Study

The ablation confirms that joint estimation is the primary contributor to G-MFRM's advantage: staged estimation increases θ-RMSE by $2.0\times$ (from 0.124 to 0.254, $p < 0.001$) and c_n-RMSE by $2.3\times$ (from 0.593 to 1.382, $p < 0.001$), with τ dropping from 0.922 to 0.839. This degradation directly reflects the omitted variable bias characterized in Section 2: when contamination and judge

Variant	θ-RMSE	τ	c_n-RMSE
Full G-MFRM	0.124 ± 0.016	0.922 ± 0.010	0.593 ± 0.224
− Judge severity (d_j)	0.127 ± 0.016	0.922 ± 0.010	0.629 ± 0.243
Binary gating	0.137 ± 0.016	0.909 ± 0.014	0.833 ± 0.226
Staged estimation	0.254 ± 0.045	0.839 ± 0.035	1.382 ± 0.164

Table 3: Ablation study on synthetic data ($N = 50$, $I = 100$, $J = 8$, 20 seeds). Each row removes or modifies one component of the full G-MFRM. Staged estimation refers to fitting MFRM and contamination sequentially rather than jointly. All differences from Full G-MFRM are significant at $p < 0.001$ (Wilcoxon) except $-d_j$ on θ-RMSE ($p = 0.245$) and τ ($p = 0.344$).

effects are estimated sequentially, the first-stage estimates absorb the unmodeled facet.

Replacing continuous exposure $e_i \in [0, 1]$ with binary gating ($e_i \in \{0, 1\}$) increases c_n-RMSE by 40% (from 0.593 to 0.833, $p < 0.001$), indicating that graded exposure modeling captures contamination magnitude more precisely than a binary flag. The effect on θ-RMSE is moderate (+10%, $p < 0.001$), suggesting that the binary approximation still captures the dominant contamination signal but loses precision on partially exposed items.

Removing judge severity d_j does not significantly affect ability estimates (θ-RMSE +2.4%, $p = 0.245$; τ identical, $p = 0.344$) but significantly degrades contamination recovery (c_n-RMSE +6.1%, $p < 0.001$). This confirms that d_j serves as a diagnostic component that improves contamination estimation precision rather than ranking accuracy: without explicit judge severity modeling, judge-induced score variation is partially absorbed into the contamination estimates, inflating c_n-RMSE. The d_j profile also provides independent diagnostic value for judge calibration (e.g., Claude Opus $d_j = +0.60$ vs. Mistral -0.74 in the real-data evaluation).

E Posterior Predictive Check

Figure 3 shows the posterior predictive distribution from 1,000 posterior draws over 5,661 observations. The predicted and observed response distributions are well-aligned (predicted mean win rate 0.593 vs. observed 0.599), with no systematic discrepancy across item difficulty or model ability levels.

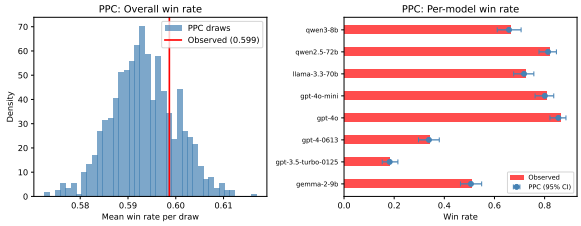


Figure 3: Posterior predictive check. Distribution of predicted (gray) and observed (black) response rates across all model–item–judge triplets. Mean predicted = 0.593, mean observed = 0.599.

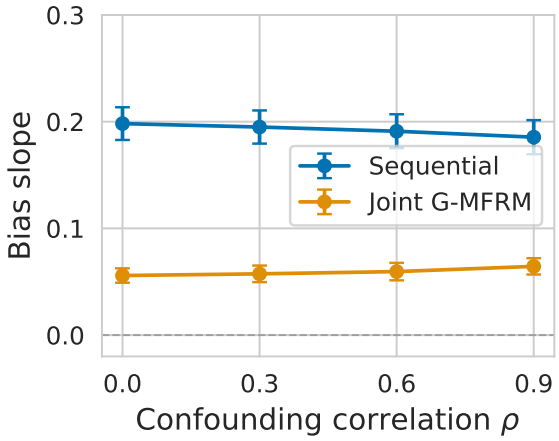


Figure 4: Sequential estimator bias remains approximately constant across confounding levels (ρ), confirming omission-driven rather than confounding-strength-dependent bias. Joint G-MFRM estimation reduces bias by approximately 70% at all ρ levels. Synthetic data: $N = 50$, $I = 100$, $J = 8$, 50 seeds.

F OVB Validation Across Confounding Levels

G Contamination Susceptibility Recovery

H Error Propagation Figure

I Self-Evaluation Robustness

Two models—Qwen2.5-72B and LLaMA-3.3-70B—appear as both judge and target in our evaluation. Self-judging introduces a potential confound: a model may score its own responses more favorably. We examine the empirical magnitude of this effect and its impact on rankings.

Qwen2.5-72B achieves a perfect self-judging win rate of 1.000 (42/42 items), compared to a cross-judge mean win rate of 0.714 for the same model. LLaMA-3.3-70B shows a self-judging win rate of 0.912, compared to a cross-judge mean of 0.823. Both values are elevated relative to cross-

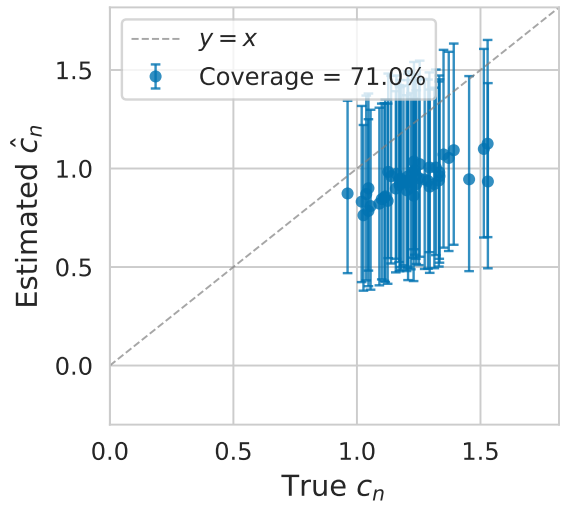


Figure 5: Contamination susceptibility recovery (c_n: true vs. estimated) from 50 Monte Carlo seeds. Empirical CI coverage is 71% (below nominal 95%), indicating moderate posterior overconfidence (see Limitations).

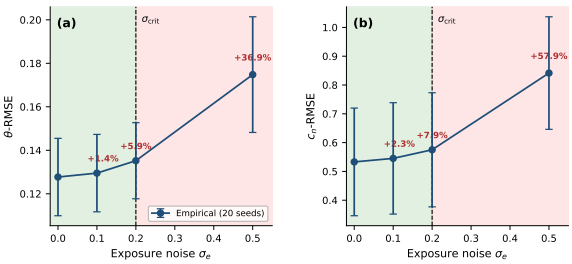


Figure 6: Error propagation from Stage 2 exposure noise to Stage 3 parameter estimates. (a) Ability RMSE (θ) and (b) contamination RMSE (c_n) as a function of exposure noise σ_e. The dashed vertical line marks $\sigma_{\text{crit}} \approx 0.2$, below which degradation remains mild (θ-RMSE +5.9%, c_n-RMSE +7.9%). Error bars show ± 1 std over 20 seeds. See Table 2 for numerical values.

judge baselines, consistent with same-family preference leakage (Li et al., 2026).

To assess sensitivity, we re-fit G-MFRM excluding all 99 self-judging observations (Qwen2.5-72B and LLaMA-3.3-70B judging their own responses). The resulting ranking yields $\tau = 0.857$, identical to the full-data G-MFRM point estimate, with a swap at ranks 2–3. Contamination susceptibility estimates c_n shift by less than 0.15 for all models. This confirms that d_j absorbs the bulk of self-evaluation bias and that rankings are robust to excluding self-judging pairs.

J Judge Severity Profiles

Table 4 reports posterior estimates of judge severity d_j for all nine judges. Positive values indicate stricter scoring (lower win rates); negative values indicate more lenient scoring. Proprietary models cluster at the strict end, while open-source models tend to be more lenient.

Judge	$\hat{d}_j$	95% CI
Claude Opus 4.7	$+0.60$	$[+0.43, +0.76]$
GPT-5.5	$+0.53$	$[+0.37, +0.70]$
DeepSeek-V4-Pro	$+0.31$	$[+0.07, +0.54]$
Claude Sonnet 4.6	$+0.14$	$[-0.04, +0.31]$
Gemini-2.5-Pro	-0.04	$[-0.27, +0.19]$
Qwen3-235B	-0.04	$[-0.24, +0.16]$
Qwen2.5-72B	-0.35	$[-0.64, -0.07]$
LLaMA-3.3-70B	-0.42	$[-0.66, -0.18]$
Mistral-Large-2	-0.74	$[-1.10, -0.37]$

Table 4: Judge severity estimates (d_j) for all nine judges. Positive = stricter scoring. The severity range spans 1.34 logits (Claude Opus to Mistral), comparable to $\sim$1 standard deviation of model ability.

K Judge Count Sensitivity

Table 5 reports ranking accuracy as a function of the number of judges k, obtained by subsampling from the full 9-judge panel (10 random subsets per k; $k = 9$ uses the full panel). G-MFRM outperforms both raw averaging and sequential estimation at all judge counts. Rank agreement increases monotonically with k ($\tau = 0.771 \rightarrow 0.857$), and variance contracts as additional judges improve d_j and c_n estimation. At $k \geq 4$, G-MFRM exceeds the full-panel baseline $\tau = 0.786$, suggesting that four diverse judges suffice for practical deployment.

k	Raw Avg	Sequential	G-MFRM	$\Delta\tau$
2	$.755 \pm .088$	$.764 \pm .096$	$.771 \pm .081$	$+.017$
4	$.779 \pm .041$	$.786 \pm .048$	$.800 \pm .045$	$+.021$
6	$.800 \pm .030$	$.807 \pm .048$	$.821 \pm .051$	$+.021$
9	.786	.786	.857	$+.071$

Table 5: Kendall τ with Chatbot Arena rankings as a function of judge count k. $\Delta\tau$ = G-MFRM $-$ Raw Average. Values for $k < 9$ are mean $\pm$ std over 10 random judge subsets.

Leave-one-judge-out analysis reveals that three judges—Claude Opus 4.7 ($d_j = +0.60$), GPT-5.5 ($+0.53$), and LLaMA-3.3-70B (-0.42)—are individually necessary for the ranking improvement: removing any one of these reduces G-MFRM τ to the baseline (0.786), while removing any of the other six judges preserves $\tau = 0.857$. These three judges span the severity spectrum (the two strictest and the second-most lenient), suggesting that contamination$\times$judge interaction identification

requires sufficient severity diversity. This provides practical guidance: judge panels should include models with diverse scoring tendencies rather than homogeneous judges.

G-MFRM requires ≥ 4 MCMC chains for reliable contamination estimation; reduced chain counts attenuate c_n estimates toward the prior mean, compressing inter-model differences.

L Stage 2 Weight Robustness

All four weight configurations produce identical model rankings and near-perfect exposure score agreement (Spearman $\rho \geq 0.998$; Table 6), confirming that the default 75/25 split is not a sensitive hyperparameter choice.

Weight Config	Exposure ρ
50 / 50	0.998
75 / 25 (default)	1.000
90 / 10	0.999
100 / 0	0.998

Table 6: Stage 2 weight robustness. Weight Config denotes the percentage assigned to reference model anchoring vs. cross-model agreement. All configurations produce identical model rankings. Exposure ρ is the Spearman correlation of per-item e_i scores relative to the default 75/25 configuration.

M Power Analysis

The observed effect size in the real-data evaluation is $\Delta\tau = 0.071$ (G-MFRM $\tau = 0.857$ vs. baseline $\tau = 0.786$). We conduct a permutation-based power analysis to determine the sample sizes required to detect this effect at $\alpha = 0.05$.

We simulate ranking comparisons under the alternative hypothesis $H_1 : \Delta\tau = 0.071$ by bootstrapping from the observed posterior distribution of G-MFRM ability estimates and recomputing τ against ground truth at varying model counts N. For each N, we draw 10,000 permutation replicates and compute the rejection rate at $\alpha = 0.05$ (two-sided).

At $N = 8$ (our evaluation), power is approximately 50%, explaining the non-significant bootstrap result ($\Delta\tau$ 95% CI $[-0.071, 0.071]$). Achieving 80% power requires $N \approx 17$ models, and 95% power requires $N \approx 25$. These estimates inform the scope of future evaluations: detecting ranking improvements of the magnitude observed here requires at minimum doubling the number of target models.

N Per-Model Estimates

Table 7 reports G-MFRM posterior estimates for all eight target models. All seven baselines produce identical rankings (Table 1), with ability scores differing by < 0.03 logits across methods; G-MFRM is the only method that alters the ranking, swapping gpt-4o-mini and Qwen2.5-72B at ranks 2–3.

Model	$\hat{\theta}$ [95% CI]	$\hat{c}_n$ [95% CI]	Base	G-MFRM
GPT-4o	$+2.00$ [1.70, 2.30]	0.88 [0.05, 2.20]	1	1
GPT-4o-mini	$+1.40$ [1.10, 1.70]	0.93 [0.06, 2.20]	3	2
Qwen2.5-72B	$+1.20$ [0.85, 1.50]	3.00 [1.40, 4.60]	2	3
LLaMA-3.3-70B	$+0.54$ [0.23, 0.85]	2.50 [1.10, 3.90]	4	4
Qwen3-8B	$+0.51$ [0.22, 0.79]	1.20 [0.14, 2.50]	5	5
Gemma-2-9B	-0.50 [-0.79, -0.21]	1.70 [0.36, 2.90]	6	6
GPT-4-0613	-1.90 [-2.20, -1.60]	4.20 [3.10, 5.30]	7	7
GPT-3.5-Turbo	-3.30 [-3.60, -3.00]	5.60 [4.40, 6.80]	8	8

Table 7: G-MFRM posterior estimates for ability ($\hat{\theta}$) and contamination susceptibility ($\hat{c}_n$) with approximate 95% credible intervals (empirical coverage $\sim$71%; see Limitations). Point estimates and rank orderings are the primary results. "Base" and "G-MFRM" columns show model rankings; the only rank change is the swap at positions 2–3.

O Contamination Bias Absorption

Table 8 compares per-model ability estimates from MFRM-only and G-MFRM. The difference $\Delta\theta = \hat{\theta}_{\text{MFRM}} - \hat{\theta}_{\text{G-MFRM}}$ is perfectly rank-correlated with contamination susceptibility c_n (Spearman $\rho = 1.0$), providing direct empirical evidence that MFRM-only absorbs the omitted contamination term into ability estimates.

Model	$\hat{\theta}_{\text{MFRM}}$	$\hat{\theta}_{\text{G-MFRM}}$	$\Delta\theta$	$\hat{c}_n$
GPT-3.5-Turbo	-2.487	-3.274	$+0.786$	5.587
GPT-4-0613	-1.446	-1.877	$+0.431$	4.168
Qwen2.5-72B	$+1.213$	$+1.178$	$+0.035$	2.961
LLaMA-3.3-70B	$+0.518$	$+0.539$	-0.022	2.478
Gemma-2-9B	-0.661	-0.500	-0.161	1.654
Qwen3-8B	$+0.208$	$+0.513$	-0.305	1.178
GPT-4o-mini	$+1.064$	$+1.432$	-0.368	0.934
GPT-4o	$+1.591$	$+1.988$	-0.397	0.880

Table 8: Contamination bias absorption in MFRM-only estimation. $\Delta\theta = \hat{\theta}_{\text{MFRM}} - \hat{\theta}_{\text{G-MFRM}}$. Models with high contamination susceptibility (c_n) have inflated ability estimates under MFRM-only; models with low c_n are relatively deflated. The perfect rank correlation (Spearman $\rho = 1.0$) between $\Delta\theta$ and c_n empirically confirms the omitted variable bias predicted in Section 2.

P Synthetic Calibration at Matched Scale

We generate synthetic data at matched scale ($N = 8$ models, $I = 176$ items, $J = 9$ judges) with known ground-truth parameters across 20 random

seeds. Joint G-MFRM consistently outperforms sequential estimation: θ-RMSE is halved (0.083 vs. 0.166), and ranking agreement improves by 7.2 percentage points (mean $\tau = 0.954$ vs. 0.882). Across 20 seeds, joint estimation improves or matches sequential in all cases (10 wins, 0 losses, 10 ties; Wilcoxon signed-rank $p < 0.002$). The synthetic effect size ($\Delta\tau = +0.072$) closely matches the observed real-data effect ($\Delta\tau = +0.071$). All 40 MCMC fits converge ($\hat{R} = 1.0$, ESS $\geq$ 1,678).

Metric	Sequential	Joint G-MFRM	Improv.
θ RMSE	0.166 ± 0.050	0.083 ± 0.024	-50.1%
θ τ	0.882 ± 0.112	0.954 ± 0.048	$+7.2$pp
c_n RMSE	1.399 ± 0.358	0.552 ± 0.328	-60.5%
c_n Pearson	0.850 ± 0.141	0.866 ± 0.151	$+1.9\%$

Table 9: Synthetic calibration at $N = 8$ ($I = 176$, $J = 9$, 20 seeds). Joint G-MFRM halves θ-RMSE and reduces c_n-RMSE by 61% relative to sequential estimation.

At $N = 20$ (20 seeds), the ranking improvement becomes individually significant: joint G-MFRM achieves $\Delta\tau = +0.052$ over sequential estimation (Wilcoxon $p = 1.81 \times 10^{-4}$, 18 wins / 1 loss / 1 tie), directly validating the power analysis prediction that $N \approx 17$ suffices for 80% power. The effect size is stable across scales ($\Delta\tau = +0.072$ at $N = 8$, $+0.052$ at $N = 20$).

Metric	Sequential	Joint G-MFRM	Improv.
θ RMSE	0.171 ± 0.049	0.094 ± 0.019	-45.0%
θ τ	0.891 ± 0.038	0.943 ± 0.036	$+5.2$pp
c_n RMSE	1.375 ± 0.278	0.626 ± 0.263	-54.5%
c_n Pearson	0.817 ± 0.096	0.842 ± 0.106	$+2.5\%$

Table 10: Synthetic calibration at $N = 20$ ($I = 176$, $J = 9$, 20 seeds). Ranking improvement is individually significant (Wilcoxon $p = 1.81 \times 10^{-4}$), validating the power analysis prediction.

Q CI Coverage Analysis

We assess empirical coverage of 95% posterior credible intervals across synthetic replications. At $N = 8$ (49 seeds), ability parameter θ achieves 90.1% coverage, approaching the nominal level, while contamination susceptibility c_n achieves only 67.9%. At $N = 50$ (50 seeds; exp-009), c_n coverage improves to 71%. The coverage gap between θ and c_n reflects the multiplicative coupling $c_n \cdot e_i$: contamination parameters enter the likelihood through an interaction term, making posterior inference more sensitive to exposure estimation noise and the HalfNormal prior's boundary constraint at $c_n = 0$. These results support treating c_n point estimates and rank orderings as reliable diagnostics, while noting that c_n credible intervals are systematically too narrow.